Week 7 – High-Performance Pandas and Apache Arrow

02613 Python and High-Performance Computing

Last week on 02613…

- Why should reduction-operators be commutative?
  – Why no matrix multiplication?

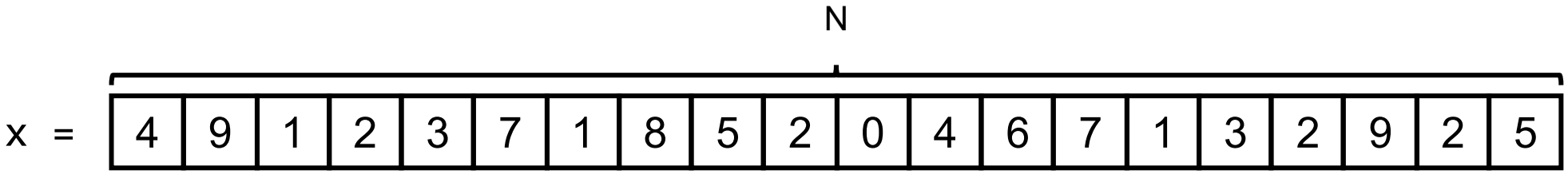
- The “simple” and binary-tree algorithms from last week do not reorder operands

- Many frameworks *require* this – allows for optimizations in implementation

- Some frameworks *support* non-commutative operators

  – May negatively impact performance – use only when needed

So far: array data



Today: tabular data

coordsx	coordsy	observed	parameterId	stationId	value
9,7875	56,1496	2023-01-17T23:59:00Z	precip_past1min	5185	0
8,1465	56,5032	2023-01-17T23:59:00Z	precip_past1min	5296	0
9,2877	57,0143	2023-01-17T23:59:00Z	precip_past1min	5085	0
8,6652	54,9872	2023-01-17T23:59:00Z	precip_past1min	5355	0
12,0981	55,2076	2023-01-17T23:59:00Z	precip_past1min	5889	0
14,7494	55,0677	2023-01-01T00:00:00Z	wind_dir_past1h	6190	209
-51,7308	64,1833	2023-01-01T00:00:00Z	temp_min_past1h	4250	-9,2
10,1272	56,3027	2023-01-01T00:00:00Z	precip_past10min	6072	0,5
9,4041	56,879	2023-01-01T00:00:00Z	precip_dur_past10min	5081	10
11,6035	55,7358	2023-01-01T00:00:00Z	precip_dur_past1h	6156	21

Today: tabular data

coordsx	coordsy	observed	parameterId	stationId	value
9,7875	56,1496	2023-01-17T23:59:00Z	precip_past1min	5185	0
8,1465	5			5296	0
9,2877	5			5085	0
8,6652	5			5355	0
12,0981	5			5889	0
14,7494	5			6190	209
-51,7308	6			4250	-9,2
10,1272	5			6072	0,5
9,4041	56,879	2023-01-01T00:00:00Z	precip_dur_past10min	5081	10
11,6035	55,7358	2023-01-01T00:00:00Z	precip_dur_past1h	6156	21

pandas

Pandas

```
>>> import pandas as pd
>>> df = pd.read_csv('data.csv')
>>> df
   coordsx  coordsy              observed           parameterId  stationId  value
0   9.7875  56.1496   2023-01-17T23:59:00Z        precip_past1min       5185    0.0
1   8.1465  56.5032   2023-01-17T23:59:00Z        precip_past1min       5296    0.0
2   9.2877  57.0143   2023-01-17T23:59:00Z        precip_past1min       5085    0.0
3   8.6652  54.9872   2023-01-17T23:59:00Z        precip_past1min       5355    0.0
4  12.0981  55.2076   2023-01-17T23:59:00Z        precip_past1min       5889    0.0
5  14.7494  55.0677   2023-01-01T00:00:00Z        wind_dir_past1h       6190  209.0
6 -51.7308  64.1833   2023-01-01T00:00:00Z        temp_min_past1h       4250   -9.2
7  10.1272  56.3027   2023-01-01T00:00:00Z       precip_past10min       6072    0.5
8   9.4041  56.8790   2023-01-01T00:00:00Z   precip_dur_past10min       5081   10.0
9  11.6035  55.7358   2023-01-01T00:00:00Z      precip_dur_past1h       6156   21.0
```

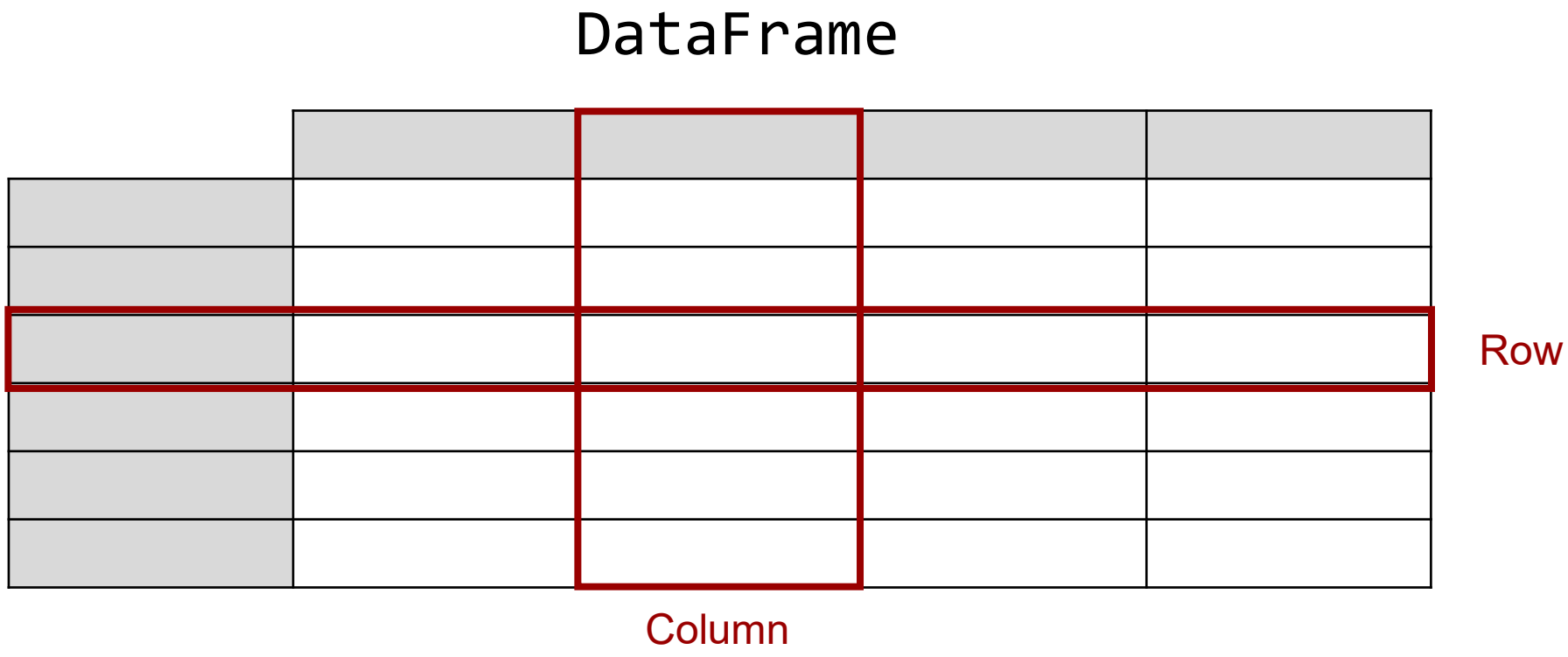
”Typical” pandas workflow

	Bottleneck
1. Load data	Load speed
2. Explore data	<u>Mem. use</u> + speed
3. Settle on workflow	Mem. use + <u>speed</u>

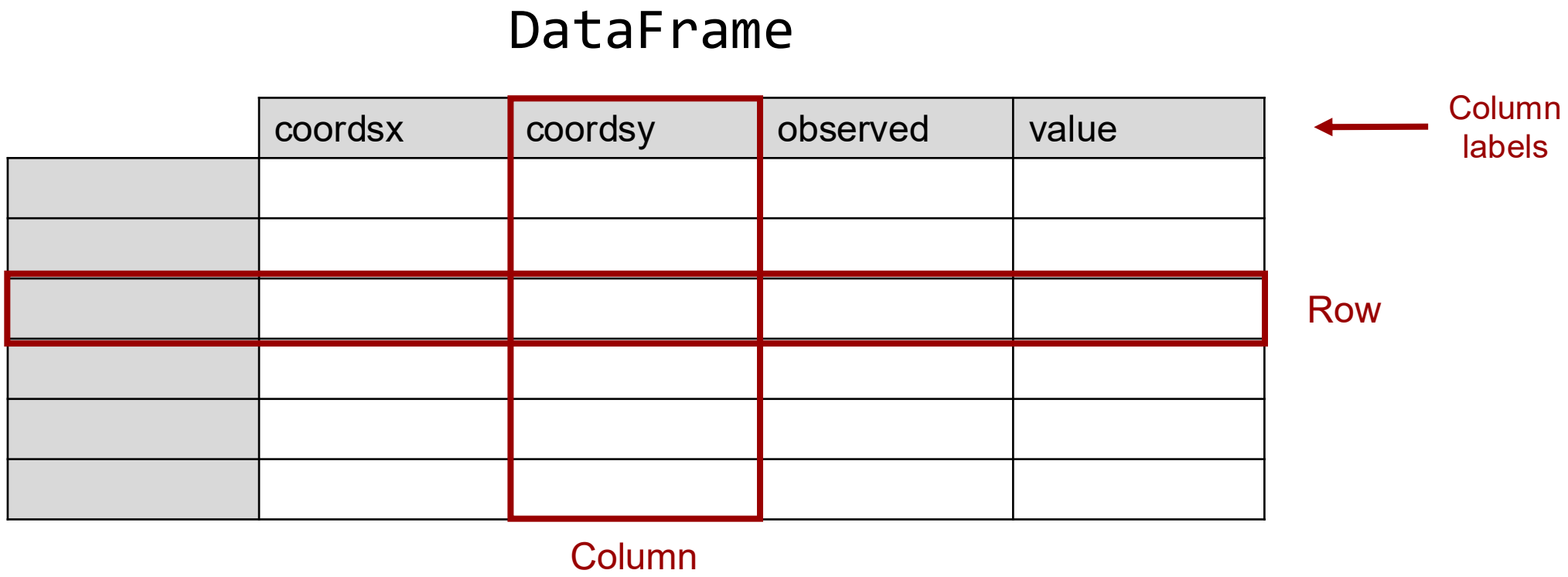
Today

- Loading data
- Minimizing memory use
- Fast operations

About pandas



About pandas



About pandas

DataFrame

	coordsx	coordsy	observed	value
0				
1				
2				
3				
4				
5				

Column labels

Row

Column

Row labels / index

About pandas

```
>>> import pandas as pd
>>> df = pd.read_csv('data.csv')
>>> df['coordsx']
0      9.7875
1      8.1465
2      9.2877
3      8.6652
4     12.0981
5     14.7494
...
Name: coordsx, dtype: float64
```

Each column has its own data type

Series

0	
1	
2	
3	
4	
5	

Data
(NumPy array)

Row labels / index

About pandas

```
>>> import pandas as pd
>>> a = pd.Series([1,2,3,4])
>>> b = pd.Series([1,2,3,4])
>>> a + b
0    2
1    4
2    6
3    8
dtype: int64
>>> import numpy as np
>>> np.mean(a)
2.5
```

a

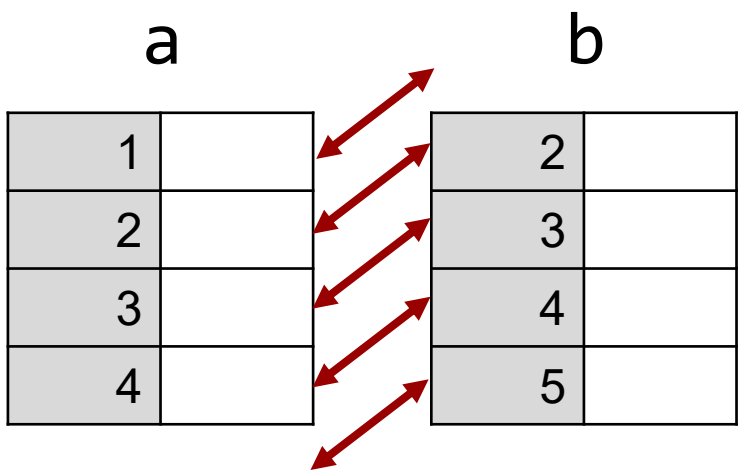
0	
1	
2	
3	

b

0	
1	
2	
3	

About pandas

```
>>> import pandas as pd
>>> a = pd.Series([1,2,3,4], index=[1,2,3,4])
>>> b = pd.Series([1,2,3,4], index=[2,3,4,5])
>>> a + b
1    NaN
2    3.0
3    5.0
4    7.0
5    NaN
dtype: float64
```



About pandas

```
>>> import pandas as pd
>>> a = pd.Series([1,2,3,4], index=[1,2,3,4])
>>> b = pd.Series([1,2,3,4], index=[2,3,4,5])
>>> a.to_numpy() + b.to_numpy()  # Views
array([2, 4, 6, 8])
```

a

1	
2	
3	
4	

b

2	
3	
4	
5	

About pandas

```
>>> import pandas as pd
>>> a = pd.Series([1,2,3,4], index=[1,2,3,4])
>>> b = pd.Series([1,2,3,4], index=[2,3,4,5])
>>> df = pd.DataFrame({ ‘a’: a, ‘b’: b })
>>> df
     a    b
1  1.0  NaN
2  2.0  1.0
3  3.0  2.0
4  4.0  3.0
5  NaN  4.0
```

df

	a	b
1		
2		
3		
4		
5		

About pandas

```
>>> import pandas as pd
>>> a = pd.Series([1,2,3,4], index=[1,2,3,4])
>>> b = pd.Series([1,2,3,4], index=[2,3,4,5])
>>> df = pd.DataFrame({ ‘a’: a, ‘b’: b })
>>> df * 2
     a    b
1  2.0  NaN
2  4.0  2.0
3  6.0  4.0
4  8.0  6.0
5  NaN  8.0
```

df

	a	b
1		
2		
3		
4		
5		

About pandas

- DataFrames are 2d tables that hold the data
- Each column is a Series with its own data type (often backed by NumPy array)
- DataFrames and Series behave like NumPy arrays
- … except for the index!

Quiz time!

Loading data

```
$ ls
bighuge.csv.zip
$ unzip bighuge.csv.zip
$ ls
bighuge.csv.zip       bighuge.csv
$ python
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv’)
```

Loading data

```
$ ls
bighuge.csv.zip
$ python
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv.zip’)
```

Faster (potentially)

Uses less disk space

Loading data

```
$ ls
bighuge.csv.zip
$ python
>>> imp
>>> df
```

More to come!

Faster (potentially)

Uses less disk space

Memory use

```
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv.zip’)
>>> df.info(memory_usage=‘deep’)
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   coordsx      10 non-null     float64
 1   coordsy      10 non-null     float64
 2   observed     10 non-null     object
 4   parameterId  10 non-null     object
 5   stationId    10 non-null     int64
 6   value        10 non-null     float64
dtypes: float64(3), int64(1), object(3)
memory usage: 2.0 GB
```

Memory use

```
>>> def summarize_columns(df):
...     print(pd.DataFrame([
...         (
...  Column name  c,
...  Data type  df[c].dtype,
...  Num. uniq. vals  len(df[c].unique()),
...  Mem. use [MB]  df[c].memory_usage(deep=True) // (1024**2)
...         ) for c in df.columns
...     ], columns=['name', 'dtype', 'unique', 'size (MB)']))
...     print('Total size:', df.memory_usage(deep=True).sum() / 1024**2, 'MB')
...
>>>
```

Total mem. use of dataframe

Memory use

```
>>> summarize_columns(df)
                    name    dtype   unique  size (MB)
0                VendorID  float64        3         48
1     tpep_pickup_datetime   object  2134342        464
2    tpep_dropoff_datetime   object  2137286        464
3          passenger_count  float64       11         48
4            trip_distance  float64     5606         48
5        store_and_fwd_flag   object        3        352
6             DOLocationID    int64      262         48
7             payment_type  float64        6         48
8              fare_amount  float64     5283         48
9                  mta_tax  float64       11         48
10              tip_amount  float64     3626         48
11            total_amount  float64    12488         48
Total size: 2014.113718032837 MB
```

Memory use

1. Date types ← Date ending with E not A

2. Recoding categorial variables

3. Smaller numeric data types

Date types

```
>>> df[‘tpep_pickup_datetime’]
0          2020-01-01 00:28:15
1          2020-01-01 00:35:39
2          2020-01-01 00:47:41
3          2020-01-01 00:55:23
4          2020-01-01 00:01:58
                 ...
6405003    2020-01-31 22:51:00
6405004    2020-01-31 22:10:00
6405005    2020-01-31 22:50:07
6405006    2020-01-31 22:25:53
6405007    2020-01-31 22:44:00
Name: tpep_pickup_datetime, Length: 6405008, dtype: object
```

Issue: datetimes are stored as <u>strings</u>

Date types

```
>>> df["tpep_pickup_datetime"] = pd.to_datetime(df["tpep_pickup_datetime"])
>>> df["tpep_dropoff_datetime"] = pd.to_datetime(df["tpep_dropoff_datetime"])
>>> df['tpep_pickup_datetime']
0         2020-01-01 00:28:15
1         2020-01-01 00:35:39
2         2020-01-01 00:47:41
3         2020-01-01 00:55:23
4         2020-01-01 00:01:58
                  ...
6405003   2020-01-31 22:51:00
6405004   2020-01-31 22:10:00
6405005   2020-01-31 22:50:07
6405006   2020-01-31 22:25:53
6405007   2020-01-31 22:44:00
Name: tpep_pickup_datetime, Length: 6405008, dtype: datetime64[ns]
```

Date types

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0                VendorID         float64        3         48
1     tpep_pickup_datetime  datetime64[ns]  2134342         48
2    tpep_dropoff_datetime  datetime64[ns]  2137286         48
3                VendorID         float64        3         48
4     tpep_pickup_datetime          object  2134342        464
5    tpep_dropoff_datetime          object  2137286        464
6          passenger_count         float64       11         48
7            payment_type         float64        6         48
8             fare_amount         float64     5283         48
9                 mta_tax         float64       11         48
10             tip_amount         float64     3626         48
11           total_amount         float64    12488         48
Total size: 1183.38600569000244 MB
```

Recoding categorical variables

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0                VendorID         float64        3         48
1     tpep_pickup_datetime  datetime64[ns]  2134342         48
2    tpep_dropoff_datetime  datetime64[ns]  2137286         48
3          passenger_count         float64       11         48
4            trip_distance         float64     5606         48
5       store_and_fwd_flag          object        3        352
6             DOLocationID           int64      262         48
7             payment_type         float64        6         48
8              fare_amount         float64     5283         48
9                  mta_tax         float64       11         48
10              tip_amount         float64     3626         48
11            total_amount         float64    12488         48
Total size: 1183.38600569000244 MB
```

Recoding categorical variables

```
>>> df['store_and_fwd_flag']
0            N
1            N
2            N
3            N
          ...
6405004    NaN
6405005    NaN
6405006    NaN
6405007    NaN
Name: store_and_fwd_flag, Length: 6405008, dtype: object
>>> df['store_and_fwd_flag'].unique()
array(['N', 'Y', nan], dtype=object)
```

Issues:

1. Missing values
2. True/False encoded as strings

Recoding categorical variables

```
>>> book_sol = df['store_and_fwd_flag'].fillna(" ").apply(ord)\
... .apply(lambda x: [32, 78, 89].index(x) - 1).astype(np.int8)
>>>
```

Recoding categorical variables

```
>>> book_sol = df['store_and_fwd_flag'].fillna(" ")
>>> book_sol.unique()
array(['N', 'Y', ' '], dtype=object)
```

Recoding categorical variables

```
>>> book_sol = df['store_and_fwd_flag'].fillna(" ").apply(ord)
>>> book_sol.unique()
array([78, 89, 32])
>>> help(ord)
Help on built-in function ord in module builtins:

ord(c, /)
    Return the Unicode code point for a one-character string.
>>> ord('a'), ord('b')
(97, 98)
```

Recoding categorical variables

```
>>> book_sol = df['store_and_fwd_flag'].fillna(" ").apply(ord)\
... .apply(lambda x: [32, 78, 89].index(x) - 1)
>>> book_sol.unique()
array([ 0,  1, -1])
```

Recoding categorical variables

```
>>> book_sol = df['store_and_fwd_flag'].fillna(" ").apply(ord)\
... .apply(lambda x: [32, 78, 89].index(x) - 1).astype(np.int8)
>>> book_sol.unique()
array([ 0,  1, -1], dtype=int8)
```

Recoding categorical variables

```
>>> book_sol = df['store_and_fwd_flag'].fillna(" ")\
... .apply(lambda x: [32, 78, 89].index(x) - 1).astype(np.int8)
>>> book_sol.unique()
array([ 0,  1, -1], dtype=int8)
>>>
>>> book_sol.memory_usage(deep=True) / 1024**2  # MB
6.108417510986328
>>>
>>> df['store_and_fwd_flag'].memory_usage(deep=True) / 1024**2  # MB
352.6583957672119
```

Recoding categorical variables

```
>>> pandas_sol = df['store_and_fwd_flag'].astype('category')
>>> pandas_sol.unique()
['N', 'Y', NaN]
Categories (2, object): ['N', 'Y']
>>> pandas_sol.memory_usage(deep=True) / 1024**2  # MB
6.108631134033203
>>>
>>> book_sol.memory_usage(deep=True) / 1024**2  # MB
6.108417510986328
>>>
>>> pandas_sol.cat.codes.unique()
array([ 0,  1, -1], dtype=int8)
```

Recoding categorical variables

- Categorical variables = can take one of a fixed limited set of values

- Recoding = assign each value to a (possibly arbitrary) number, then replace values with numbers

- Can be done manually or with pandas ‘category’ type

Recoding categorical variables

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0                VendorID         float64        3         48
1     tpep_pickup_datetime  datetime64[ns]  2134342         48
2    tpep_dropoff_datetime  datetime64[ns]  2137286         48
3          passenger_count         float64       11         48
4            trip_distance         float64     5606         48
5        store_and_fwd_flag        category        3          6
6             DOLocationID           int64      262         48
7             payment_type         float64        6         48
8              fare_amount         float64     5283         48
9                  mta_tax         float64       11         48
10              tip_amount         float64     3626         48
11            total_amount         float64    12488         48
Total size: 836.8362922668457 MB
```

Numeric data types

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0                VendorID         float64        3         48
1     tpep_pickup_datetime  datetime64[ns]  2134342         48
2    tpep_dropoff_datetime  datetime64[ns]  2137286         48
3          passenger_count         float64       11         48
4            trip_distance         float64     5606         48
5        store_and_fwd_flag        category        3          6
6             DOLocationID           int64      262         48
7             payment_type         float64        6         48
8              fare_amount         float64     5283         48
9                  mta_tax         float64       11         48
10              tip_amount         float64     3626         48
11            total_amount         float64    12488         48
Total size: 836.8362922668457 MB
```

Numeric data types

```
>>> df["payment_type"].unique()
array([ 1.,  2.,  4.,  3.,  5., nan])
>>> df["payment_type"] = df["payment_type"].fillna(0.0).astype(np.int8)
```

Numeric data types

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0               VendorID         float64        3         48
1    tpep_pickup_datetime  datetime64[ns]  2134342         48
2   tpep_dropoff_datetime  datetime64[ns]  2137286         48
3        passenger_count         float64       11         48
4          trip_distance         float64     5606         48
5      store_and_fwd_flag        category        3          6
6           DOLocationID           int64      262         48
7           payment_type            int8        6          6
8            fare_amount         float64     5283         48
9                mta_tax         float64       11         48
10            tip_amount         float64     3626         48
11          total_amount         float64    12488         48
Total size: 794.0782508850098 MB
```

Numeric data types

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0               VendorID         float64        3         48
1    tpep_pickup_datetime  datetime64[ns]  2134342         48
2   tpep_dropoff_datetime  datetime64[ns]  2137286         48
3                                                 11         48
4
5
6           DOLocationID           int64
7           payment_type            int8        6          6
8            fare_amount         float64     5283         48
9                mta_tax         float64       11         48
10             tip_amount         float64     3626         48
11           total_amount         float64    12488         48
Total size: 794.0782508850098 MB
```

When does this work?

Numeric data types

Integers (whole numbers)

int8
int16
int32
int64

Unsigned integers (non-negative)

uint8
uint16
uint32
uint64

Floating point (fractional numbers)

float16
float32
float64

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767

```
>>> df["payment_type"].unique()
array([ 1.,  2.,  4.,  3.,  5., nan])
>>> df["payment_type"] = df["payment_type"].fillna(0.0).astype(np.int8)
```

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Key point: can convert to smaller integer when:

1. Values are whole numbers
2. Min/max fits the data type

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767
int32	-2,147,483,648	2,147,483,647
int64	-9.22 * 10^18	9.22 * 10^18

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

np.finfo

Floating point resolution

$$123.4$$

Exponent

$$1.234 \cdot 10^2$$

Significant / mantissa (only has one digit left of the point)

Resolution ≈ nr. of digits in significant

Floating point resolution

$$123.4$$

Exponent

Note: on the computer everything is in base 2!

Resolution ≈ nr. of digits in significant

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$123.4$$

Exponent

$$1.234 \cdot 10^2$$

Significant / mantissa

Resolution ≈ nr. of digits in significant

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$12340.0$$

Exponent

$$1.234 \cdot 10^4$$

Significant / mantissa

Resolution ≈ nr. of digits in significant

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$0.0001234$$

Exponent

$$1.234 \cdot 10^{-4}$$

Significant / mantissa

Resolution ≈ nr. of digits in significant

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$1.0 + 0.0001234$$

Exponent

$$?.???\cdot 10^{?}$$

Significant / mantissa

Resolution ≈ nr. of digits in significant

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$1.0 + 0.0001234$$

$$1.000 \cdot 10^{0} + 1.234 \cdot 10^{-4}$$

$$1.0001234 \cdot 10^{0}$$

Can only keep these

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$1.0 + 0.0001234$$

$$1.000 \cdot 10^{0} + 1.234 \cdot 10^{-4}$$

$$1.000 \cdot 10^{0}$$

Rounding error

Resolution = 0.001 (like float16)

Floating point resolution

Example:
4 digits for sig
Exponent fro

$$1.0 + 0.0001234$$

Note: resolution is relative to the larger number

Rounding error
Resolution = 0.001 (like float16)

Floating point resolution

Example:
4 digits for significant
Exponent from -4 to 4

$$10.0 + 0.001234$$

$$1.000 \cdot 10^{1} + 1.234 \cdot 10^{-3}$$

$$1.0001234 \cdot 10^{1}$$

Can only keep these

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767
int32	-2,147,483,648	2,147,483,647
int64	-9.22 * 10^18	9.22 * 10^18

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

```
>>> a = np.array(1.0, dtype='float16')
>>> b = np.array(0.1, dtype='float16')
>>> +a, +b  # Unary + makes it print nice
(1.0, 0.1)
>>> a + b
1.1
```

np.finfo

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767
int32	-2,147,483,648	2,147,483,647
int64	-9.22 * 10^18	9.22 * 10^18

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

```
>>> a = np.array(1.0, dtype='float16')
>>> b = np.array(0.001, dtype='float16')
>>> +a, +b  # Unary + makes it print nice
(1.0, 0.001)
>>> a + b
1.001
```

np.finfo

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767
int32	-2,147,483,648	2,147,483,647
int64	-9.22 * 10^18	9.22 * 10^18

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

```
>>> a = np.array(1.0, dtype='float16')
>>> b = np.array(0.0001, dtype='float16')
>>> +a, +b  # Unary + makes it print nice
(1.0, 0.0001)
>>> a + b
1.0
```

Rounding error!

np.finfo

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767
int32	-2,147,483,648	2,147,483,647
int64	-9.22 * 10^18	9.22 * 10^18

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

```
>>> a = np.array(1.0, dtype='float16')
>>> b = np.array(0.001, dtype='float16')
>>> +a, +b  # Unary + makes it print nice
(1.0, 0.001)
>>> a + b
1.001
```

np.finfo

Numeric data types

Integers (whole numbers)

	min	max
int8	-128	127
int16	-32,768	32,767
int32	-2,147,483,648	2,147,483,647
int64	-9.22 * 10^18	9.22 * 10^18

Unsigned integers (non-negative)

uint8	0	255
uint16	0	65535
uint32	0	4,294,967,295
uint64	0	1.84 * 10^19

np.iinfo

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

```
>>> a = np.array(10.0, dtype='float16')
>>> b = np.array(0.001, dtype='float16')
>>> +a, +b  # Unary + makes it print nice
(10.0, 0.001)
>>> a + b
10.0
```

Rounding error!

np.finfo

Numeric data types

```
>>> summarize_columns(df)
                    name           dtype   unique  size (MB)
0                VendorID         float64        3         48
1     tpep_pickup_datetime  datetime64[ns]  2134342         48
2    tpep_dropoff_datetime  datetime64[ns]  2137286         48
3          passenger_count         float64       11         48
4            trip_distance         float64     5606         48
5        store_and_fwd_flag        category        3          6
6             DOLocationID           int64      262         48
7             payment_type            int8        6          6
8              fare_amount         float64     5283         48
9                  mta_tax         float64       11         48
10              tip_amount         float64     3626         48
11            total_amount         float64    12488         48
Total size: 794.0782508850098 MB
```

Numeric data types

```
>>> df['fare_amount']
0           6.00
1           7.00
2           6.00
3           5.50
4           3.50
           ...
6405003    17.59
6405004    46.67
6405005    48.85
6405006    27.17
6405007    54.56
Name: fare_amount, Length: 6405008, dtype: float64
```

Numeric data types

```
>>> np.abs(df['fare_amount'] - df['fare_amount'].astype('float32')).sum()
0.06289832668632866
>>> np.abs(df['fare_amount'] - df['fare_amount'].astype('float32')).max()
2.4414062522737368e-05
>>>
>>> np.abs(df['fare_amount'] - df['fare_amount'].astype('float16')).sum()
511.4022631836173
>>> np.abs(df['fare_amount'] - df['fare_amount'].astype('float16')).max()
1.0
```

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

Numeric data types

```
>>> np.abs(df['fare_amount'] - df['fare_amount'].astype('float32')).sum()
0.06289832668632866
>>> np.abs(df['fare_amount'] - df['
2.4414062522737368e-05
>>>
>>> np.abs(df['fare_amount'] - df['
511.4022631836173
>>> np.abs(df['fare_amount'] - df['fare_amount'].astype('float16')).max()
1.0
```

Key point: must check for errors before using smaller float type.

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	±6.55 * 10^4
float32	1e-6	±3.40 * 10^38
float64	1e-15	±1.80 * 10^308

Memory use

1. Date types

2. Recoding categorial variables

3. Smaller numeric data types

Faster operations – row wise computations

```
def get_tip_mean_explicit(df):
    all_tips = 0
    for i in range(len(df)):  # len(df) = num. rows
        row = df.iloc[i]      # Access row by index
        all_tips += row["tip_amount"] / row["total_amount"]

    return all_tips / len(df)

t = time()
get_tip_mean_explicit(df)
print(time() - t)
# LOOOOOOOONG (minutes)
```

Slow but simple

Faster operations – row wise computations

```
def get_tip_mean_apply(df):
    frac_tip = df.apply(
        lambda row: row["tip_amount"] / row["total_amount"],
        axis=1  # Apply to each row, i.e., apply along columns (axis 1)
    )
    return frac_tip.mean()  # NumPy like syntax


t = time()
get_tip_mean_apply(df)
print(time() - t)
# 9.5 s
```

Fast-ish and general

Faster operations – row wise computations

```
def get_tip_mean_vectorized(df):
    # Divide columns with NumPy-like syntax
    frac_tip = df["tip_amount"] / df["total_amount"]
    return frac_tip.mean()


t = time()
get_tip_mean_vectorized(df)
print(time() - t)
# 32 ms
```

Fastest

Faster operations – row wise computations

```
def get_tip_mean_numpy(df):
    tip_amount_np = df["tip_amount"].to_numpy()
    total_amount_np = df["total_amount"].to_numpy()
    # Pure NumPy from here on
    frac_tip = tip_amount_np / total_amount_np
    return frac_tip.mean()


t = time()
get_tip_mean_numpy(df)
print(time() - t)
# 11 ms
```

Fastester!

Faster operations – using the index

DataFrame

	coordsx	coordsy	observed	value
0				
1				
2				
3				
4				
5				

Column labels

Row labels / index

Faster operations – using the index

DataFrame

	coordsx	coordsy	value
observed			
2023-01-17T23:59:00Z			
2023-01-17T23:59:00Z			
2023-01-17T23:59:00Z			
2023-01-17T23:59:00Z			
2023-01-17T23:59:00Z			
2023-01-01T00:00:00Z			

Column labels

Row labels / index

Faster operations – using the index

```
>>> df[  # Who paid congestion surcharge at given time \
...     (df["tpep_pickup_datetime"] == "2020-01-06 08:13:00") & \
...     (df["congestion_surcharge"] > 0)
... ]
>>>
```

Boolean indexing

Faster operations – using the index

```
>>> df[  # Who paid congestion surcharge at given time \
...     (df["tpep_pickup_datetime"] == "2020-01-06 08:13:00") & \
...     (df["congestion_surcharge"] > 0)
... ]
>>>
```

35 ms

Faster operations – using the index

```
>>> df[  # Who paid congestion surcharge at given time \
...     (df["tpep_pickup_datetime"] == "2020-01-06 08:13:00") & \
...     (df["congestion_surcharge"] > 0)
... ]
>>>
>>> df_indexed = df.set_index(“tpep_pickup_datetime”)
>>>
>>> df_time = df_indexed.loc["2020-01-06 08:13:00"]
>>> df_time[df_time["congestion_surcharge"] > 0]
>>>
```

35 ms

9.3 ms

Faster operations – using the index

```
>>> df[  # Who paid congestion surcharge at given time \
...     (df["tpep_pickup_datetime"] == "2020-01-06 08:13:00") & \
...     (df["congestion_surcharge"] > 0)
... ]
>>>
>>> df_indexed = df.set_index(“tpep_pickup_datetime”)
>>>
>>> df_time = df_indexed.loc["2020-01-06 08:13:00"]
>>> df_time[df_time["congestion_surcharge"] > 0]
>>>
>>> df_indexed_sorted = df_indexed.sort_index()
>>>
>>> df_time = df_indexed_sorted.loc["2020-01-06 08:13:00"]
>>> df_time[df_time["congestion_surcharge"] > 0]
>>>
```

35 ms

9.3 ms

597 µs

Faster operations – using the index

```
>>> df[  # Who paid congestion surcharge at given time \
...     (df["tpep_pickup_datetime"] == "2020-01-06 08:13:00") & \
...     (df[
... ]
>>>
>>> df_index
>>>
>>> df_time
>>> df_time[df_time["congestion_surcharge"] > 0]
>>>
>>> df_indexed_sorted = df_indexed.sort_index()
>>>
>>> df_time = df_indexed_sorted.loc["2020-01-06 08:13:00"]
>>> df_time[df_time["congestion_surcharge"] > 0]
>>>
```

35 ms

Key point: indexing significantly speeds up queries.
<u>Price:</u> we must first build the index

9.3 ms

597 µs

Quiz time!

Loading data

```
$ ls
bighuge.csv.zip
$ python
>>> imp
>>> df
```

More to come!

Faster (potentially)

Uses less disk space

PyArrow for faster loading

APACHE
ARROW

PyArrow for faster loading

```
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv’)      # 11.3 s
>>>
>>> from pyarrow import csv
>>> table = csv.read_csv(‘bighuge.csv’)  #  4.6 s
```

PyArrow for faster loading

```
>>> df
...
 0   VendorID                float64
 1   tpep_pickup_datetime    object
 2   tpep_dropoff_datetime   object
 3   passenger_count         float64
 4   trip_distance           float64
 5   RatecodeID              float64
 6   store_and_fwd_flag      object
 7   PULocationID            int64
 8   DOLocationID            int64
 9   payment_type            float64
 10  fare_amount             float64
 11  extra                   float64
...
```

```
>>> table
pyarrow.Table
VendorID: int64
tpep_pickup_datetime: timestamp[s]
tpep_dropoff_datetime: timestamp[s]
passenger_count: int64
trip_distance: double
RatecodeID: int64
store_and_fwd_flag: string
PULocationID: int64
DOLocationID: int64
payment_type: int64
fare_amount: double
extra: double
...
```

PyArrow for faster loading

```
>>> df
...
 0   Vend
 1   tpep
 2   tpep
 3   pass
 4   trip
 5   Rate
 6   stor
 7   PULo
 8   DOLo
 9   paym
 10  fare
 11  extr
...
```

```
>>> table
pyarrow.Table

tamp[s]
stamp[s]

...
```

Key points:

1. PyArrow handles more data types (e.g., dates) automatically.
2. PyArrow has specialized data structures for handling missing data

PyArrow for faster loading

```
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv’)      # 11.3 s
>>>
>>> from pyarrow import csv
>>> table = csv.read_csv(‘bighuge.csv’)  #  4.6 s
>>>
>>> df2 = table.to_pandas()
```

PyArrow for faster loading

```
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv’)      # 11.3 s
>>>
>>> from pyarrow import csv
>>> table = csv.read_csv(‘bighuge.csv’)  #  4.6 s
>>>
>>> import pyarrow as pa
>>> copts = csv.ConvertOptions(
...     column_types={
...         "VendorID": pa.bool_(),
...         ”store_and_fwd_flag”: pa.dictionary(pa.int32(), pa.string()),
...     }
... )
>>> table = csv.read_csv(‘bighuge.csv’, convert_options=copts)
```

PyArrow for faster loading

```
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv’)      # 11.3 s
>>>
>>> from pyarrow import csv
>>> table = csv.read_csv(‘bighuge.csv’)  #  4.6 s
```

See more in the book and the official documentation

```
...         "VendorID": pa.bool_(),
...         ”store_and_fwd_flag”: pa.dictionary(pa.int32(), pa.string()),
...     }
... )
>>> table = csv.read_csv(‘bighuge.csv’, convert_options=copts)
```

Parquet files for even faster loading

```
1,2020-01-01 00:28:15,2020-01-01 00:33:03,1,1.20,1,N,238,239,1,6,3,0.5,1.47,0,0.3,11.27,2.5
1,2020-01-01 00:35:39,2020-01-01 00:43:04,1,1.20,1,N,239,238,1,7,3,0.5,1.5,0,0.3,12.3,2.5
1,2020-01-01 00:47:41,2020-01-01 00:53:52,1,.60,1,N,238,238,1,6,3,0.5,1,0,0.3,10.8,2.5
1,2020-01-01 00:55:23,2020-01-01 01:00:14,1,.80,1,N,238,151,1,5.5,0.5,0.5,1.36,0,0.3,8.16,0
2,2020-01-01 00:01:58,2020-01-01 00:04:16,1,.00,1,N,193,193,2,3.5,0.5,0.5,0,0,0.3,4.8,0
2,2020-
2,2020-
2,2019-
2,2019-
1,2020-
1,2020-
1,2020-
1,2020-01-01 00:56:27,2020-01-01 01:21:44,1,3.30,1,N,161,144,1,17,3,0.5,4.15,0,0.3,24.95,2.5
2,2020-01-01 00:21:54,2020-01-01 00:27:31,1,1.07,1,N,43,239,1,6,0.5,0.5,1.96,0,0.3,11.76,2.5
2,2020-01-01 00:38:01,2020-01-01 01:15:21,1,7.76,1,N,143,25,1,28.5,0.5,0.5,4.84,0,0.3,37.14,2.5
1,2020-01-01 00:15:35,2020-01-01 00:27:06,3,1.60,1,N,211,234,2,9,3,0.5,0,0,0.3,12.8,2.5
1,2020-01-01 00:41:20,2020-01-01 00:44:22,1,.50,1,Y,234,90,1,4,3,0.5,1,0,0.3,8.8,2.5
1,2020-01-01 00:56:38,2020-01-01 01:13:34,1,1.70,1,N,246,142,2,11.5,3,0.5,0,0,0.3,15.3,2.5
2,2020-01-01 00:08:21,2020-01-01 00:25:29,1,8.45,1,N,138,216,2,24.5,0.5,0.5,0,0,0.3,25.8,0
1,2020-01-01 00:25:39,2020-01-01 00:27:05,1,.00,1,N,170,162,4,3,3,0.5,0,0,0.3,6.8,2.5
```

Issues:

- Text wastes space when storing data
- Text requires parsing when reading / writing

Parquet files for even faster loading

```
>>> import pyarrow as pa
>>> from pyarrow import csv
>>> import pyarrow.parquet as pq
>>>
>>> table = csv.read_csv('bighuge.csv')  # 4.6 s
>>>
>>> pq.write_table(table, 'bighuge.parquet')
>>>
>>> pq.read_table('bighuge.parquet')     # 719 ms

$ ls -lh
-rw-r--r-- 1 patmjen s14 567M Dec 26  2022 bighuge.csv
-rw-r--r-- 1 patmjen s14 111M Jan 26 13:52 bighuge.csv.zip
-rw-r--r-- 1 patmjen s14 108M Mar 12 18:11 bighuge.parquet
```

Parquet files for even faster loading

```
>>> import pyarrow as pa
>>> from pyarrow import csv
>>> import pyarrow.parquet as pq
>>>
>>> table
>>>
>>> pq.wr
>>>
>>> pq.read_table('bighuge.parquet')     # 719 ms

$ ls -lh
-rw-r--r-- 1 patmjen s14 567M Dec 26  2022 bighuge.csv
-rw-r--r-- 1 patmjen s14 111M Jan 26 13:52 bighuge.csv.zip
-rw-r--r-- 1 patmjen s14 108M Mar 12 18:11 bighuge.parquet
```

Key point: Binary parquet files are much smaller and much faster

Quiz time!

Today’s exercise

Pandas and PyArrow

- Reading files with pandas
- Reading files with PyArrow
- Reading files with Parquet
- Vectorized and indexed operations

3. Check the columns and list all the changes you can do to save memory. Hint: check sections 7.1.2 and 7.1.3 on Fast Python for inspiration. You can use the following function summarize the dataframe:

```
def summarize_columns(df):
print(pd.DataFrame([
    (
        c,
        df[c].dtype,
        len(df[c].unique()),
        df[c].memory_usage(deep=True) // (1024**2)
    ) for c in df.columns
], columns=['name', 'dtype', 'unique', 'size (MB)']))
print('Total size:', df.memory_usage(deep=True).sum() / 1024**2, 'MB')
```

Useful concepts

Read with pandas

```
import pandas as pd
df = pd.read_csv('data.csv')
```

Read with PyArrow

```
from pyarrow import csv
table = csv.read_csv('data.csv')

df = table.to_pandas()  # Convert back to pandas
```

Read with Parquet

```
import pyarrow.parquet as pq
table = pq.read_table('data.csv')
```

Change to work node

linuxsh

Submit job script

bsub < submit.sh

Job status

bstat / bjobs

Check job output

bpeek / bpeek <JOBID>

Kill job

bkill <JOBID>